

JAMES PATRICK WHALEN

Senior Software Engineer

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Summary

James is an experienced **Senior Software Engineer** with a strong background in developing scalable **front-end** applications and robust **back-end** systems as well as **Machine Learning**, **Database Design**. He has a proven track record of leading projects that utilize cutting-edge technologies like **React**, **Vue**, **Node.js**, **Golang**, and **Python**. James has extensive experience in cloud computing, including **AWS**, **Docker**, **Kubernetes** and **GCP**, and excels in creating dynamic user interfaces, high-performance APIs, and implementing **microservice** architectures. He is adept at optimizing SQL and NoSQL database performance and using **CI/CD** tools for automated deployment. In **Agile** environments, James has successfully led cross-functional teams, mentored junior developers, and ensured the delivery of high-quality, innovative solutions, leveraging **Agile methodologies** to enhance project efficiency and adaptability.

Experience

Fortunes Tech

Staff Software Engineer

Goffstown, NH

05/2021 - Present

IT Services and IT Consulting

- Led and participated in the development of various web and mobile apps including **Lemonade**, **Thinknum**, and **AELER** using cutting-edge technologies, including **React**, **React Native**, **Next.js**, **Node.js**, **Nest**, **Golang**, **Python**, and **Ruby on Rails**.
- Implemented high-performance **REST APIs** endpoints for **AELER** and **Thinknum** using **Node.js**, **Nest**, **PostgreSQL**, **Supabase** and **Prisma**.
- Developed **pixel-perfect** web and mobile frontend of **Thinknum** and **Lemonade** which processes visual data and model using **TypeScript**, **Next**, **React Query**, **Recoil**, **Shadcn**, **Styled Component** and **Tailwind CSS**.
- Utilized **RabbitMQ** in **AELER** project to manage and streamline **real-time** messaging and communication between **microservices**, enhancing the efficiency and reliability of data processing and transportation logistics.
- Developed features for **Lemonade** app using **Flutter**, implementing policy management, claims tracking, and user notifications to enhance functionality and user experience on **iOS** and **Android** platforms.
- Developed an **AI chatbot** for **Lemonade** admin panel by integrating **Dialogflow** and **OpenAI** within the **Nest.js** backend service, enhancing administrative efficiency and support capabilities.
- Strong background in cloud services, including **AWS**, **GCP**, **Docker** and **Kubernetes**, proficient in designing and implementing cloud architectures, deploying and managing applications.
- Led a team of 7 members while performing **code reviews**, **scrum planning**, **mentoring**, and providing **technical leadership and guidance**.

J3st Gaming

Senior Software Engineer

Eden Prairie, MN

10/2019 - 04/2021

Online Sports Betting & Casino, operating in the UK & Ireland

- Built a web application from scratch for streamers and viewers to manage challenges and donations, integrating with **Streamlabs** and **OBS** using **JavaScript**, **Angular**, **Angular Material**, **NgRx**, **SCSS**, **HTML**, **Node.js**, and **DigitalOcean**.
- Led the migration of an **Angular** project to **Next** and deployed it on **Vercel**, enhancing **SEO** with built-in optimization features, implemented **SSR** and **SSG** for faster load times and better performance, improving user experience and content delivery.
- Transitioned UI components from **Angular Material** to **TailwindCSS**, **Radix UI** and **Styled Components**, achieving a 40% reduction in component loading speed and providing a modern and consistent look and feel.
- Enhanced the **Nest.js** backend service architecture by updating **RESTful APIs** to **GraphQL** and integrating **AWS SQS**, managing application performance effectively and improving data handling.
- Worked with a comprehensive tech stacks, including **Firebase**, **GraphQL**, **Apollo Server/Client**, **TypeScript**, **NextAuth**, **Redux**, **Zod**, **Yup**, **React Query**, **tRPC** and **microservices architecture**.
- Led the creation of a **Web3**-based sports betting pages integrating **blockchain** technologies, using **Ethereum smart contract**, **Solidity**, and **Web3.js** for secure transactions and betting logic. Enhanced user experience with **NextJS** and **Node.js** for seamless interaction.
- Developed and executed **unit testing**, **integration testing**, and **end-to-end testing** using **Jest**, **Cypress**, and **Mocha+Chai**, ensuring high-quality code and application performance while collaborating closely with the **Quality Assurance (QA)** team for comprehensive test coverage.
- Leveraged **AWS** services such as **EC2**, **S3**, **Lambda**, **RDS**, and **SQS** for deployment and optimization, ensuring robust, scalable, and high-performance application infrastructure.
- **Mentored** junior team members in advanced development principles and best practices, fostering their growth and improving team efficiency.

Kin Insurance

Software Engineer

Chicago, IL

04/2016 - 09/2019

Insurance

- Led the design and implementation of homeowners, property and life insurance underwriting software, which streamlined risk assessment processes and reduced underwriting time by 30%, enhancing client satisfaction and throughput.
- Spearheaded the implementation of API integration with multiple insurance carriers, enhancing quick quoting features that reduced quote generation time by 30% for agents.
- Collaborated with data engineering teams to develop an AI-driven tool using satellite imagery analysis, improving wildfire risk assessments by 40% and supporting underwriting decisions and implementing a powerful search feature using **Elasticsearch** for real-time.
- Led the development and integration of online signature capabilities using Dropbox Sign, achieving a 35% increase in document processing speed and enhancing customer experience.
- Implemented a seamless integration with Ivans to automate data exchanges, which enhanced data accuracy by 20% and reduced manual data entry by 50%.
- **Tech stack** - **API Integration**, **SSO**, **OKTA**, **Auth0**, **Node**, **Nest**, **TypeScript**, **Next**, **Restful API**, **Relational Databases**, **Postgres**, **Supabase**, **Sequelize**, **Training Programs**, **Build Tools**, **Stripe**, **PayPal**, **Docker**, **Kubernetes**, **Shell Scripting**.

Experience

Cengage Group

Software Engineer

Boston, MA

04/2014 - 05/2016

Software Development

- Collaborated with a cross-functional team to design and develop an **education** website and mobile application from scratch, utilizing a tech stack that includes **Python, Django, PostgreSQL, SQLAlchemy, React.js, Vue.js** and **React Native** and deployed the entire solution on **AWS**.
- Designed and implemented **RESTful APIs** using **Python, Django**, and **Django Rest Framework**, ensuring robust and scalable back-end services, utilizing **SQLAlchemy** for efficient **ORM (Object-Relational Mapping)** and database interactions with **PostgreSQL**.
- Designed a scalable **micro-service** architecture for front-end, ensured seamless **RESTful API** communication, utilized **Docker containers** for deployment, used **Nginx** for routing and load balancing, and set up **CI/CD** pipelines with **Jenkins** for automated testing and deployment.
- Developed dynamic and **responsive** user interfaces using **React, ES6, React Router** for navigation, **Redux** for state management, **MUI** for design components, **Webpack** for bundling, **Axios** for HTTP requests, **ESLint** for code quality, and **Babel** for cross-browser compatibility.
- Built admin panel components using **Vue** with two-way data binding, **Vue Router** for routing, **Vuetify** for design, and **Gulp** for automation.
- Developed a mobile app for **iOS** and **Android** with push notifications for timely updates, offline access for uninterrupted learning, real-time updates for dynamic content, interactive quizzes, progress tracking, and a user-friendly interface to enhance the overall **learning** experience.
- Deployed the application on **AWS Services** using **EC2, CloudFront, S3, SNS, Lambda, API Gateway**, and **RDS**, implementing a **Serverless Architecture** with **CI/CD pipelines** via **CodePipeline** and **CodeDeploy**, and monitoring with **CloudWatch** for performance and reliability.

Redox

Full Stack Engineer

Madison, WI

04/2012 - 03/2014

Global digital health company powering hospital at home virtual care, DCTs and digital companion apps

- Developed and maintained a health web application, migrating the existing web application from the **LAMP** stack app to **MERN** stack app.
- Translated redesigned **Figma** designs to **pixel-perfect** and fully **responsive** web application using **JavaScript, React, SASS** and **LESS**.
- Rebuilt backend services in **Node.js** using **Express.js**, creating **RESTful APIs** to handle various insurance operations such as policy management, claims processing, and user management.
- Migrated data from **MySQL** to **MongoDB**, optimizing data models and queries to take full advantage of **MongoDB's** performance capabilities.
- Implemented secure authentication flow using **JWT (JSON Web Tokens)** to ensure safe and efficient user authentication and authorization.
- Integrated **third-party services** and **APIs** to provide additional functionality, such as **payment** processing and **real-time** notifications using **Stripe, PayPal** and **WebSockets** technologies.
- Set up Continuous Integration/Continuous Deployment pipelines (**CI/CD**) using **GitHub Actions** to automate testing and deployment, ensuring rapid and reliable delivery of new features and updates.

Apexon

Software Developer

Santa Clara, CA

09/2010 - 03/2012

Software Development

- Transformed design mock-ups into **HTML5/CSS3** layouts and created **Responsive Web** libraries by utilizing the main styling tools and stuff and enhancing **UI/UX Design** in **agile development** team.
- Participated in development of **Ecommerce** and **CRM** systems with **CMS** systems including **Drupal, WordPress, Joomla** and **OOP, MVC** structure, **Laravel, SQL, Git** and followed **Pixel Perfect** design and implementation.
- Involved in writing application level code to interact with **APIs**, Web Services using **AJAX, JSON** and **XML**.
- Used **jQuery** for basic animation and end user screen customization purposes.
- Collaborated closely with senior developers to troubleshoot bugs, perform **code review** and **debugging**, and implement performance enhancements, contributing to the app's reliability, stability, and overall efficiency.

Education

George Mason University

M.S. in Computer Science

Glasgow, UK

04/2006 - 08/2010

Skills

Full Stack Development • Mobile Development • API Development • Android Development • iOS Development • Devops Engineering

Javascript • Typescript • Python • PHP • Java • Golang • Ruby • HTML5 • CSS • NPM • Yarn

ReactJS • React Native • Flutter • VueJS • Nuxt • Angular • NextJS • Gatsby • Redux • Recoil • Mobx • Zustand • Zod • Yup

SCSS • LESS • Material UI • Ant Design • Chakra UI • Styled Component • Radix UI • Shadcn • Bootstrap

Node.js • ExpressJS • NestJS • Laravel • Ruby on Rails • Django • Flask • Fast API • MySQL • PostgreSQL • Redis • MongoDB • Supabase • Firebase

Github • Gitlab • Jira • Bitbucket • CI/CD Pipeline • TDD • Jest • Cypress • Mocha+Chai • Unit Testing • E2E Testing • Jenkins

Amazon Web Services • Google Cloud Platform • Docker • Kubernetes • Microsoft Azure • Terraform • Linux • RabbitMQ • Design Patterns